Modelos en Grafos Probabilísticos (Redes Bayesianas)

Valero Laparra

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Ejemplos
Probabilidad
Probabilidad Bayesiana
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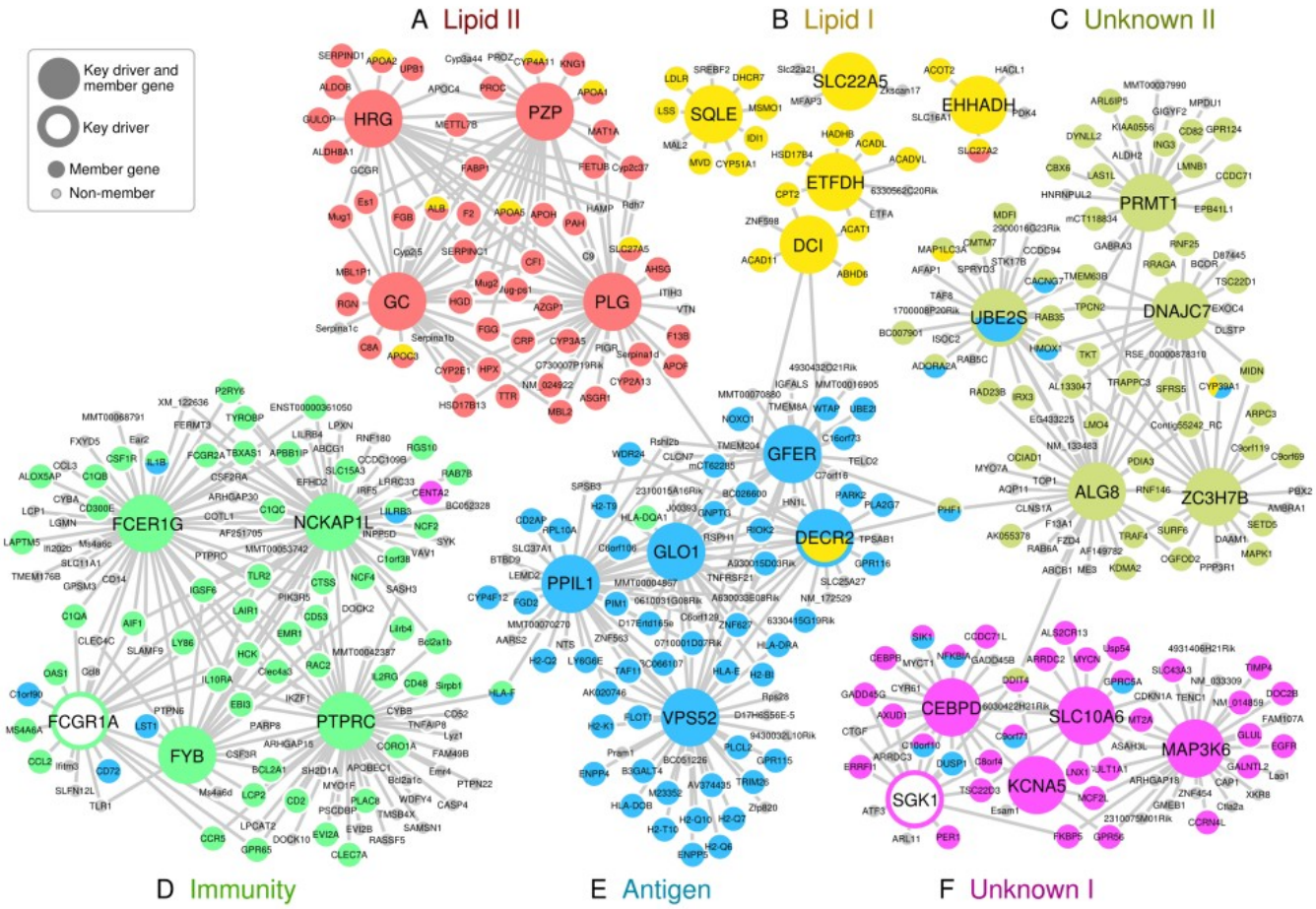
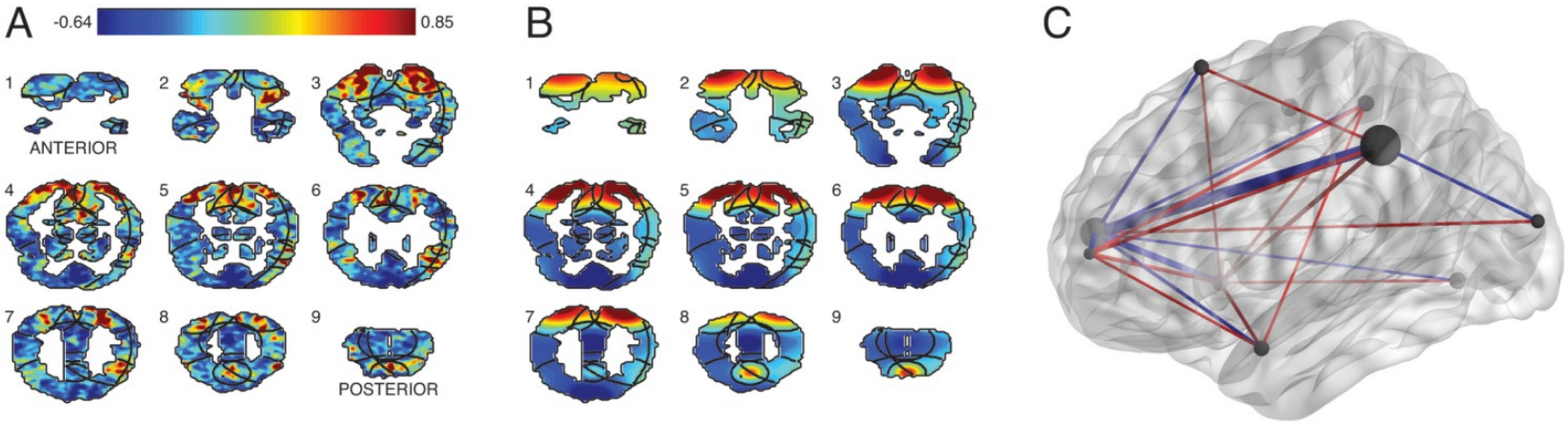
Notación

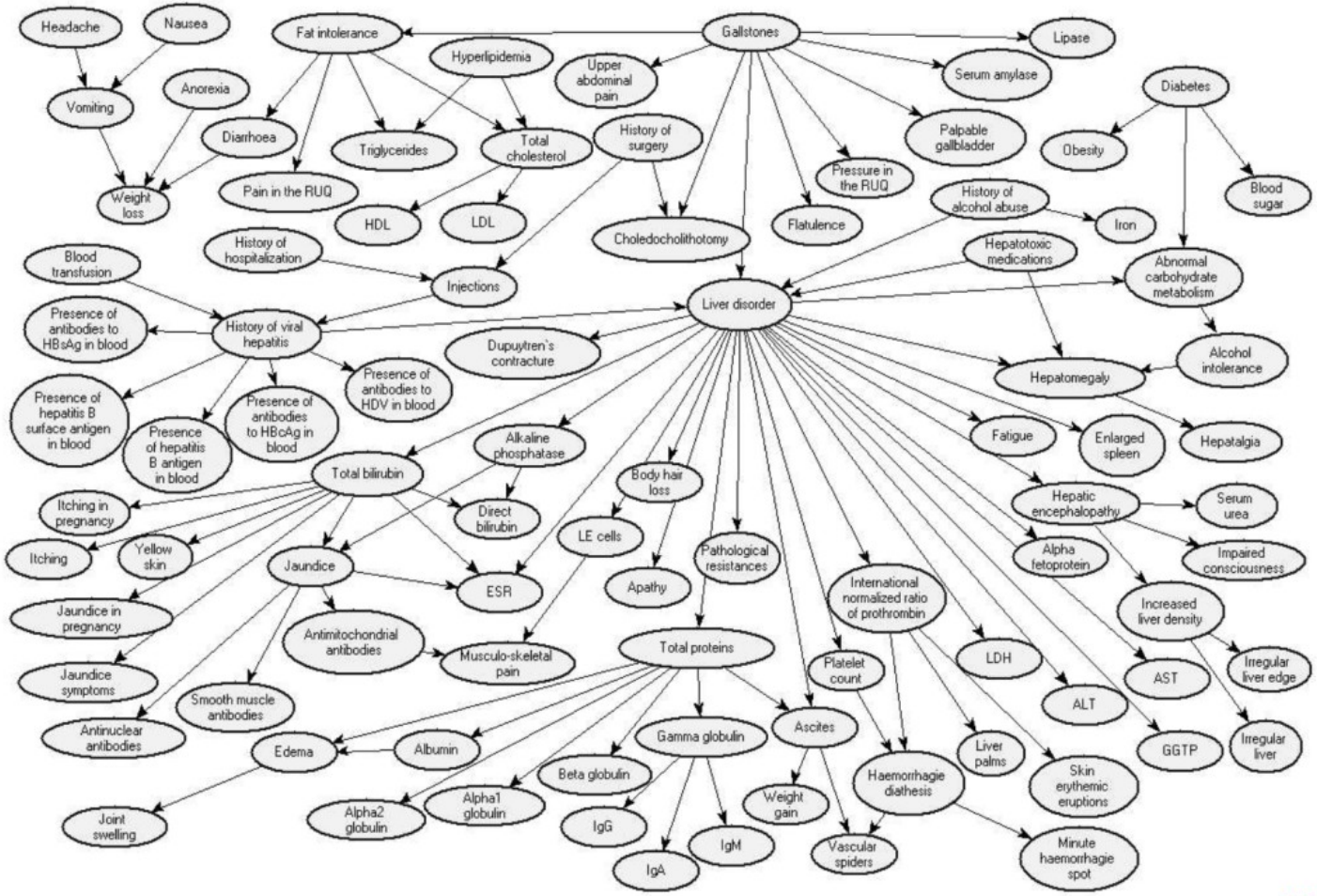


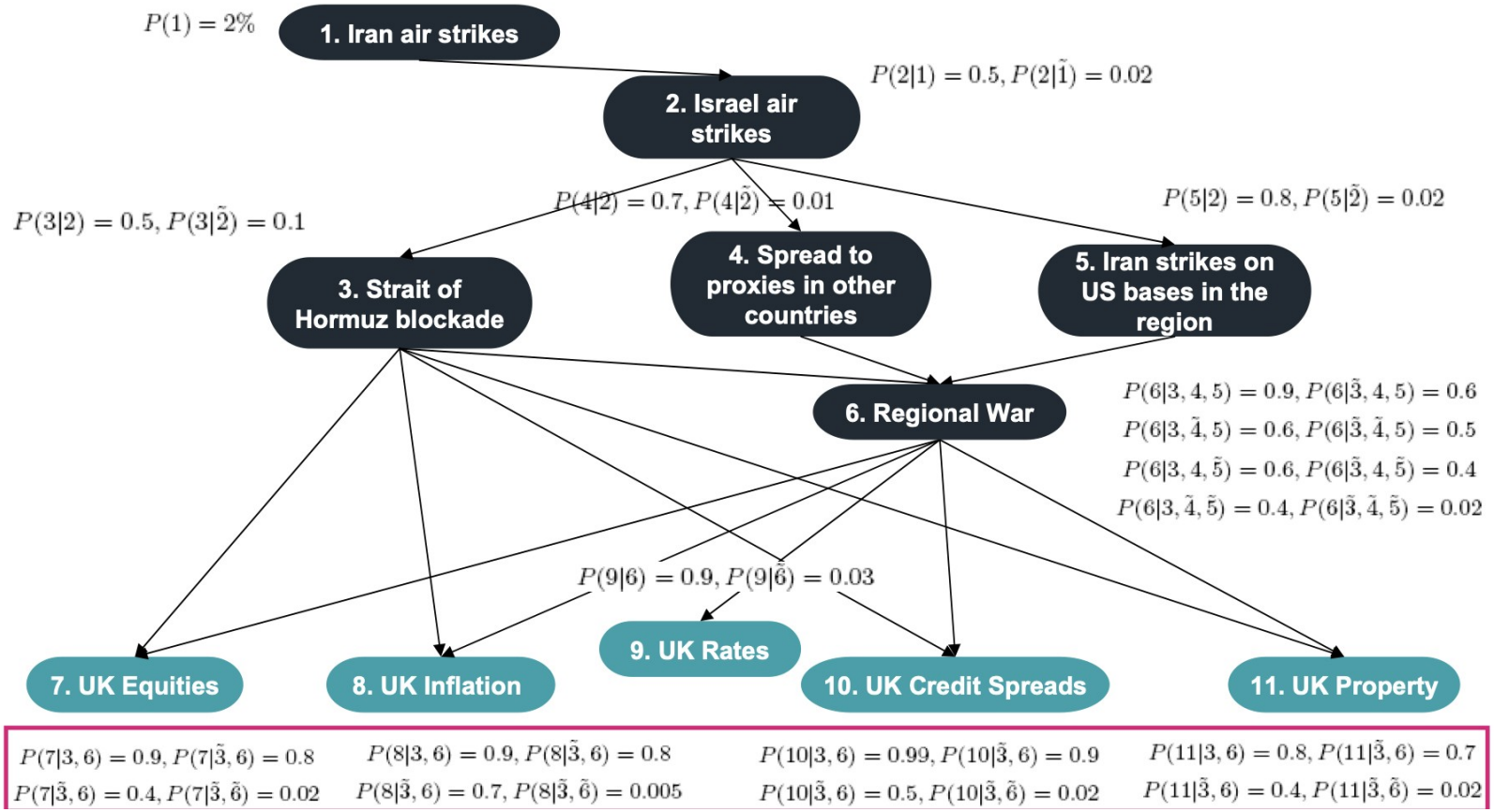
Mäkinen et al. Integrative Genomics Reveals Novel Molecular Pathways and Gene Networks for Coronary Artery Disease, PloS genetics, 2014.



Manning et al. Topographic Factor Analysis: A Bayesian Model for Inferring Brain Networks from Neural Data, PloS one, 2014.

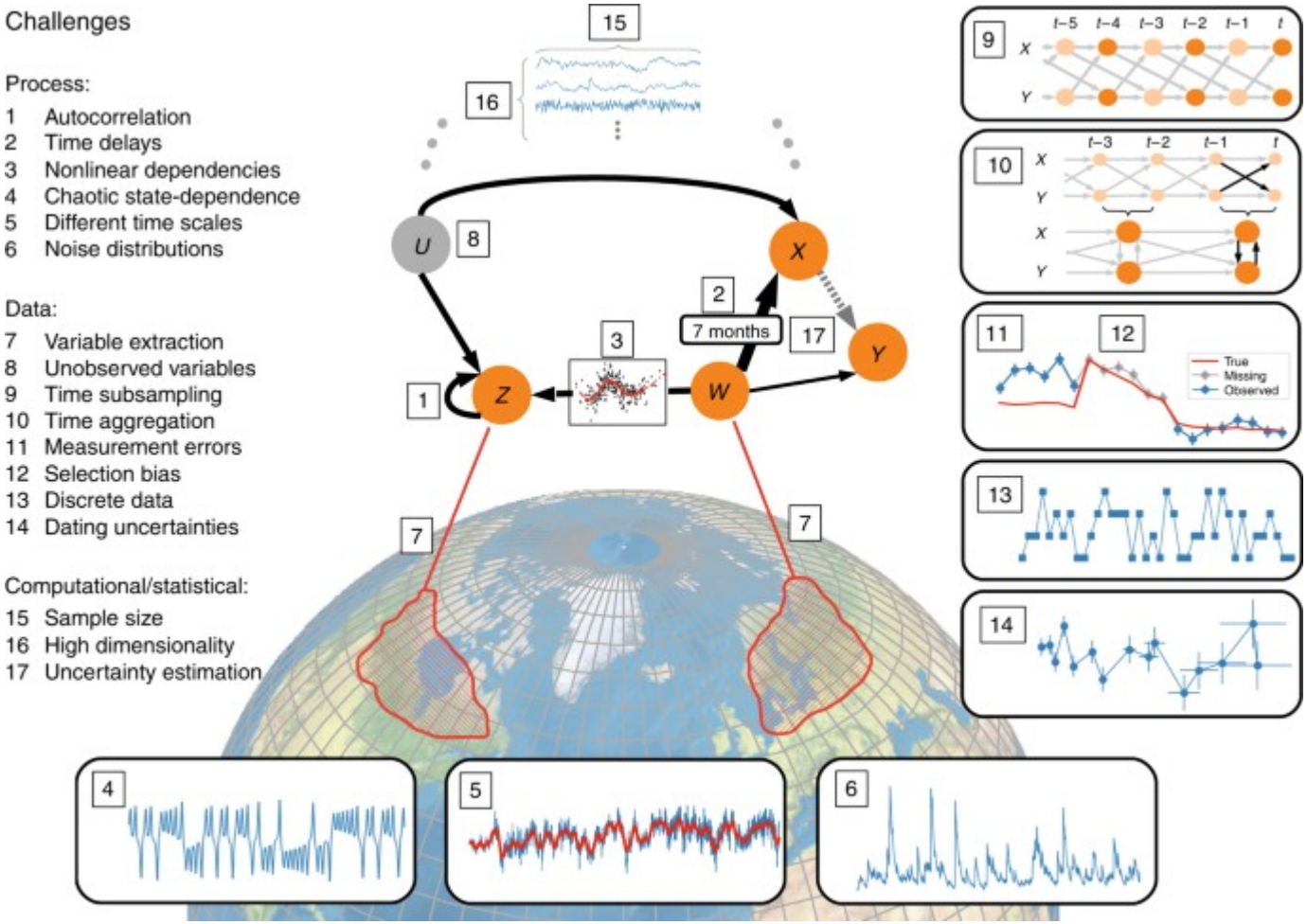
Bayesian Network for Diagnosis Liver Disorder





Probabilistic Graphical Models -- A New Way of Thinking in Financial Modelling. **Alexander Denev**

RESUMEN: https://gmarti.gitlab.io/qfin/2020/08/30/book-probabilistic-graphical-models-finance-denev.html



Inferring causation from time series in Earth system sciences. J. Runge et al. 2019

Probabilidad

Probabilidad

¿Que és?

‘Probability is the chance that a given event will occur”

— Webster dictionary

‘Probability is simply an average value’

‘Probability theory is useful to design and analyze signal processing systems’

— Gardner, 1989

“Es una medida del grado de certidumbre de que dicho suceso pueda ocurrir.”

— Wikipedia

Probabilidad

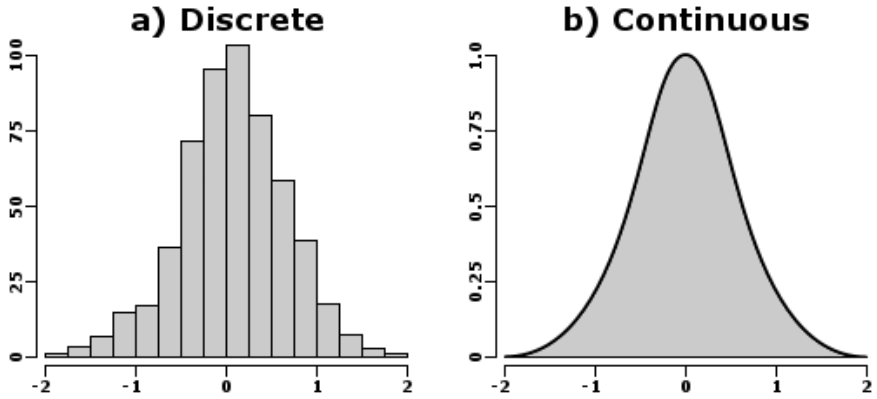
¿Que és?

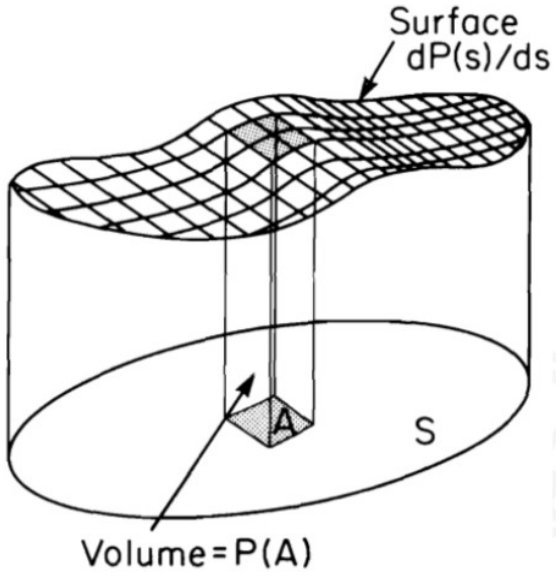
Nos da información sobre procesos que no podemos predecir

Está en TODOS sitios.

Nosotros, nuestras decisiones, las predicciones, el ruido...

Discreta / Continua

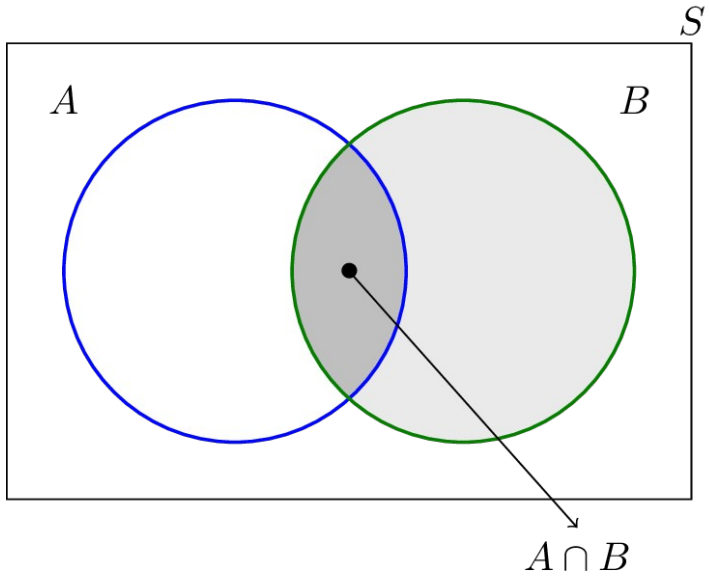




$$\Pr[a \leq X \leq b] = \int_a^b f_X(x)\,dx$$

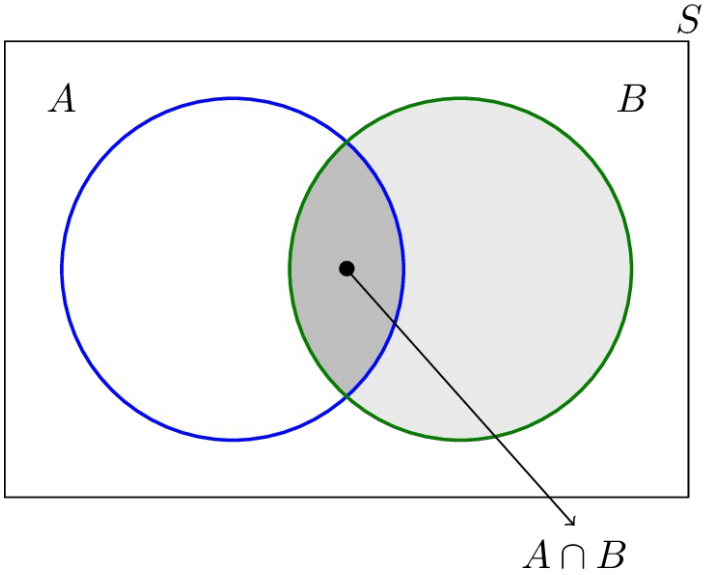
Probabilidad Bayesiana

Probabilidad condicional



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probabilidad condicional



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

Bayes

Likelihood

Prior

Posterior

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Normalizing constant

$$P(B) = \sum_{Y} P(B|A)P(A)$$

Ejercicios Bayes: Tests Covid

- Probabilidad de enfermo si positivo 1 test, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, no prior

Ejercicios Bayes: Tests Covid

· Probabilidad de enfermo si positivo 1 test, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, no prior

$p(e|+) = p(+|e)p(e) / p(+)$

Ejercicios Bayes: Tests Covid

- Probabilidad de enfermo si positivo 1 test, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, no prior

$P(+) = p(+|e)p(e) + p(+|ne)p(ne)$

Ejercicios Bayes: Tests Covid

· Probabilidad de enfermo si positivo 1 test, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, no prior

$P(+) = p(+|e)p(e) + p(+|ne)p(ne)$

$= 0.95*0.5 + 0.01*0.5 = 0.48$

Ejercicios Bayes: Tests Covid

- Probabilidad de enfermo si positivo 1 test, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, no prior

$p(e|+) = p(+|e)p(e) / p(+) = 0.95*0.5/0.48 = 0.989$

Ejercicios Bayes: Tests Covid

- Probabilidad de enfermo si positivo 1 test, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, $p(e) = 1\%$ (incidencia)

Ejercicios Bayes: Tests Covid

- Probabilidad de enfermo si positivo 1 test, si:

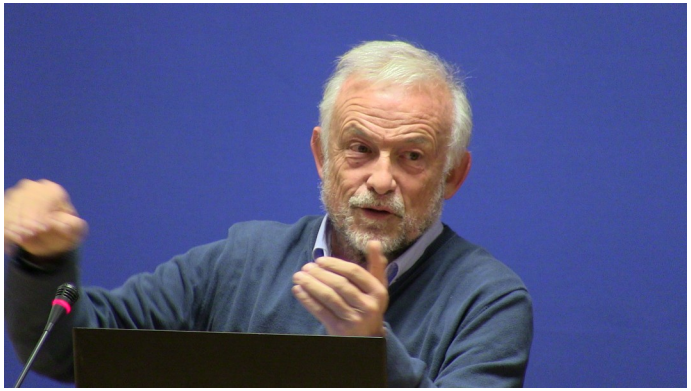
$p(+|e) = 0.95$, $p(+|ne) = 0.01$, $p(e) = 20\%$ (incidencia)

Ejercicios Bayes: Tests Covid

- Probabilidad de enfermo si positivo en 2 tests, si:

$p(+|e) = 0.95$, $p(+|ne) = 0.01$, $p(e) = 1\%$ (incidencia)

Prior no informativo



Jose Miguel Bernardo
https://www.uv.es/bernardo/

$$p = \frac{N+1}{N+2}$$

Modelos Probabilísticos

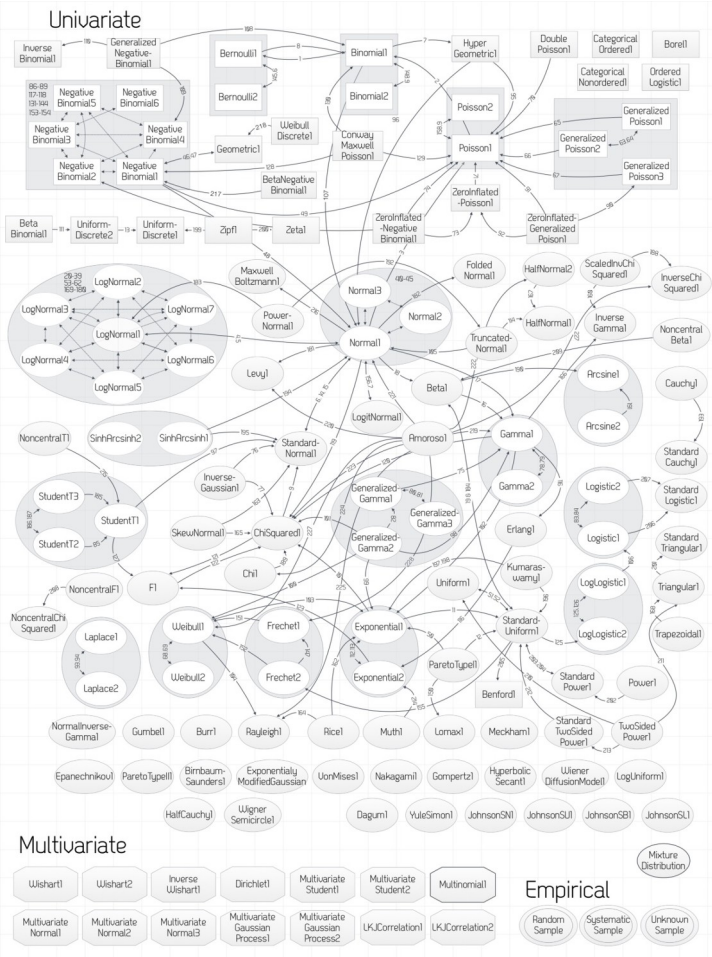
Modelos Probabilísticos

List of distributions Interactive Chart

http://www.math.wm.edu/~leemis/chart/UDR/UDR.html

List of distributions

https://en.wikipedia.org/wiki/List_of_probability_distributions



Combinar

Sumas (Convoluciones)

Si se suman dos o mas variables aleatorias independientes, la distribución de la variable resultado es la convolución de las distribuciones de las variables aleatorias

$$Y = X_1 + X_2 \qquad\qquad p_Y = p_{x1} * p_{x2}$$

Combinar

Sumas (Convoluciones)

Continuous distributions [edit]

- $$\sum_{i=1}^{n} \mathrm{Stable}(\alpha, \beta_i, c_i, \mu_i) = \mathrm{Stable}\left(\alpha, \frac{\sum_{i=1}^{n} \beta_i c_i^{\alpha}}{\sum_{i=1}^{n} c_i^{\alpha}}, \left(\sum_{i=1}^{n} c_i^{\alpha}\right)^{1/\alpha}, \sum_{i=1}^{n} \mu_i\right)$$

  $$0 < \alpha_i \leq 2 \quad -1 \leq \beta_i \leq 1 \quad c_i > 0 \quad \infty < \mu_i < \infty$$

The following three statements are special cases of the above statement:

- $$\sum_{i=1}^{n} \mathrm{Normal}(\mu_i, \sigma_i^2) \sim \mathrm{Normal}\left(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2\right) \qquad -\infty < \mu_i < \infty \quad \sigma_i^2 > 0 \quad (\alpha = 2, \beta_i = 0)$$
- $$\sum_{i=1}^{n} \mathrm{Cauchy}(a_i, \gamma_i) \sim \mathrm{Cauchy}\left(\sum_{i=1}^{n} a_i, \sum_{i=1}^{n} \gamma_i\right) \qquad -\infty < a_i < \infty \quad \gamma_i > 0 \quad (\alpha = 1, \beta_i = 0)$$

https://en.wikipedia.org/wiki/List_of_convolutions_of_probability_distributions

Evaluar/Optimizar

Likelihood, o non-negative log-likelihood (NNL)

$$\mathcal{L}_n(\theta) = \mathcal{L}_n(\theta; \mathbf{y}) = f_n(\mathbf{y}; \theta)$$

$$f_n(\mathbf{y}; \theta) = \prod_{k=1}^{n} f_k^{\text{univar}}(y_k; \theta)$$

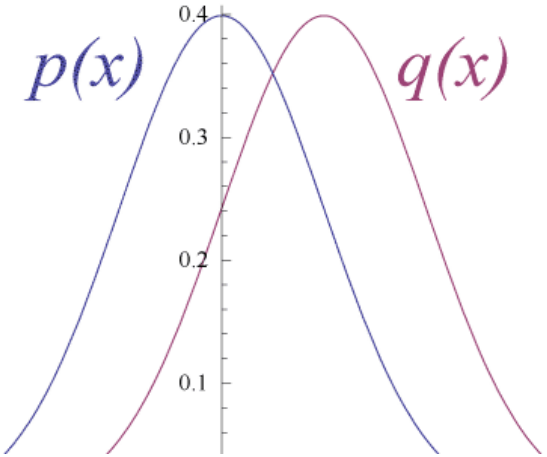
$$\hat{\theta} = \arg\max_{\theta \in \Theta} \mathcal{L}_n(\theta\,; \mathbf{y})$$

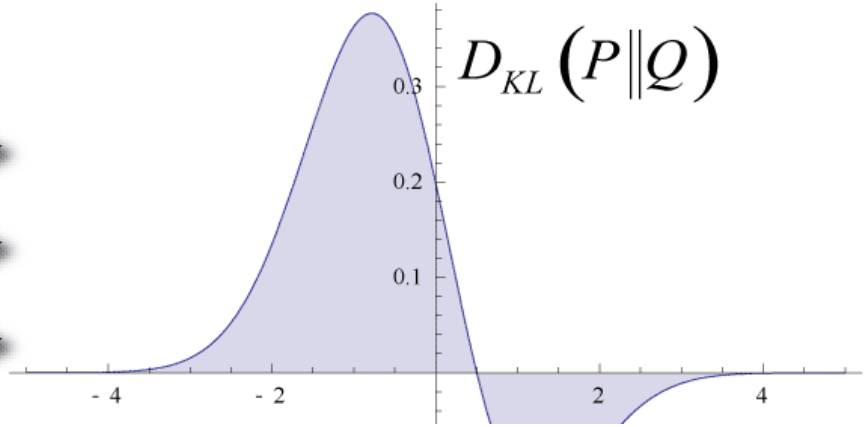
Maximum Likelihood

Evaluar/Optimizar

Divergencia de Kullback-Leibler

$$D_{\mathrm{KL}}(P\|Q) = \sum_i P(i) \ln \frac{P(i)}{Q(i)}.$$





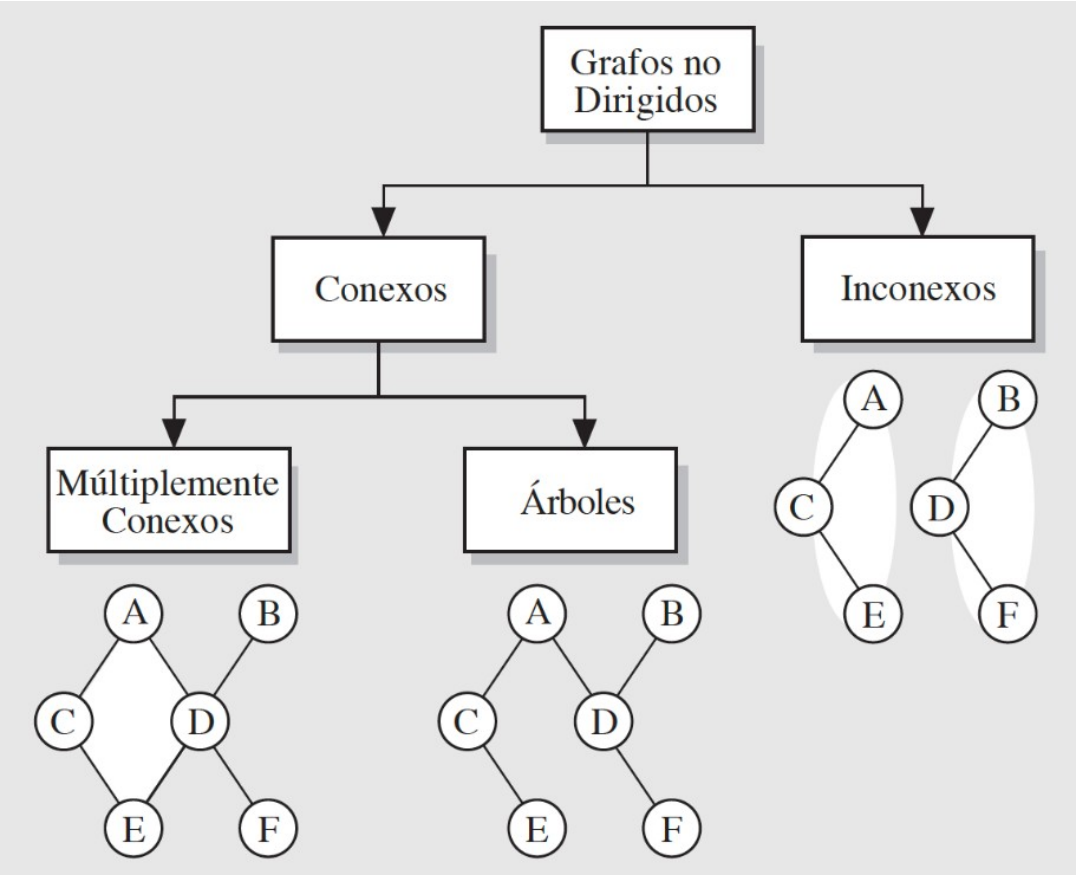
Relación max-L y min- KLD

$$
\begin{aligned}
\hat{\theta} &= \arg\max_{\theta} L_{P_\theta}(\mathbf{y}) = \arg\max_{\theta} P_\theta(\mathbf{y}) = \arg\max_{\theta} P(\mathbf{y} \mid \theta) \\
&= \arg\max_{\theta} \prod_{i=1}^{n} P(y_i \mid \theta) = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y_i \mid \theta) \\
&= \arg\max_{\theta} \left(\sum_{i=1}^{n} \log P(y_i \mid \theta) - \sum_{i=1}^{n} \log P(y_i \mid \theta_0) \right) = \arg\max_{\theta} \sum_{i=1}^{n} (\log P(y_i \mid \theta) - \log P(y_i \mid \theta_0)) \\
&= \arg\max_{\theta} \sum_{i=1}^{n} \log \frac{P(y_i \mid \theta)}{P(y_i \mid \theta_0)} = \arg\min_{\theta} \sum_{i=1}^{n} \log \frac{P(y_i \mid \theta_0)}{P(y_i \mid \theta)} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \log \frac{P(y_i \mid \theta_0)}{P(y_i \mid \theta)} \\
&= \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} h_\theta(y_i) \quad \underset{n\to\infty}{\longrightarrow} \quad \arg\min_{\theta} E[h_\theta(y)] \\
&= \arg\min_{\theta} \int P_{\theta_0}(y) h_\theta(y) dy = \arg\min_{\theta} \int P_{\theta_0}(y) \log \frac{P(y \mid \theta_0)}{P(y \mid \theta)} dy \\
&= \arg\min_{\theta} D_{\mathrm{KL}}(P_{\theta_0} \parallel P_\theta)
\end{aligned}
$$

EXTRA: Evaluar modelos generativos

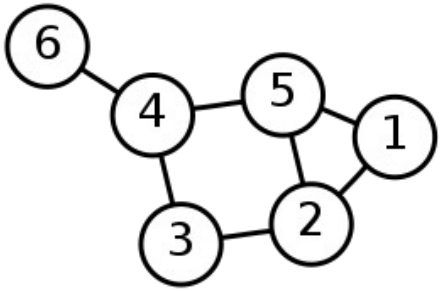
Grafos

Grafos

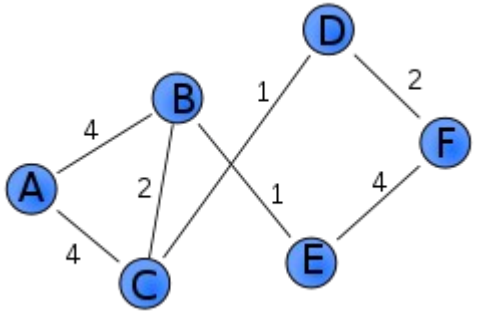

Grafos no Dirigidos
Conexos
Inconexos
Múltiplemente Conexos
Árboles
A
B
C
D
E
F

Matriz de adyacencia

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$



$$\begin{bmatrix} 0 & 4 & 4 & 0 & 0 & 0 \\ 4 & 0 & 2 & 0 & 1 & 0 \\ 4 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 & 4 & 0 \end{bmatrix}$$



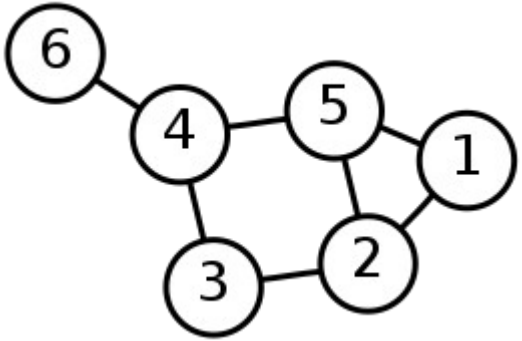
Matriz Laplaciana

$$L = D - A$$

Adyacencia

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

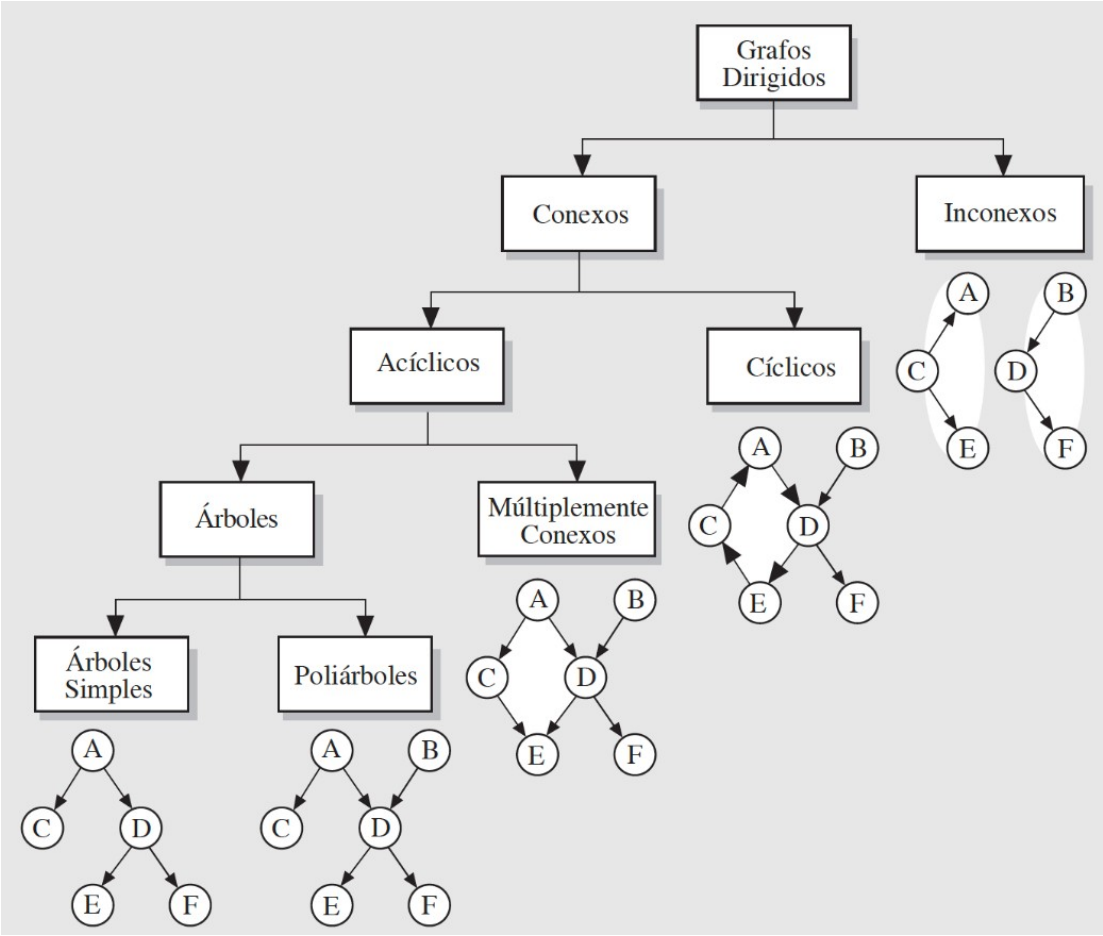
Grafo



Laplaciana

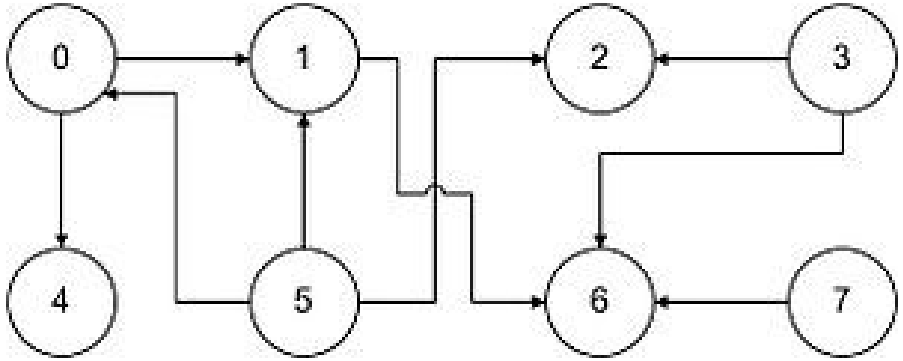
$$\begin{pmatrix} 2 & -1 & 0 & 0 & -1 & 0 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & -1 \\ -1 & -1 & 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix}$$

Grafos

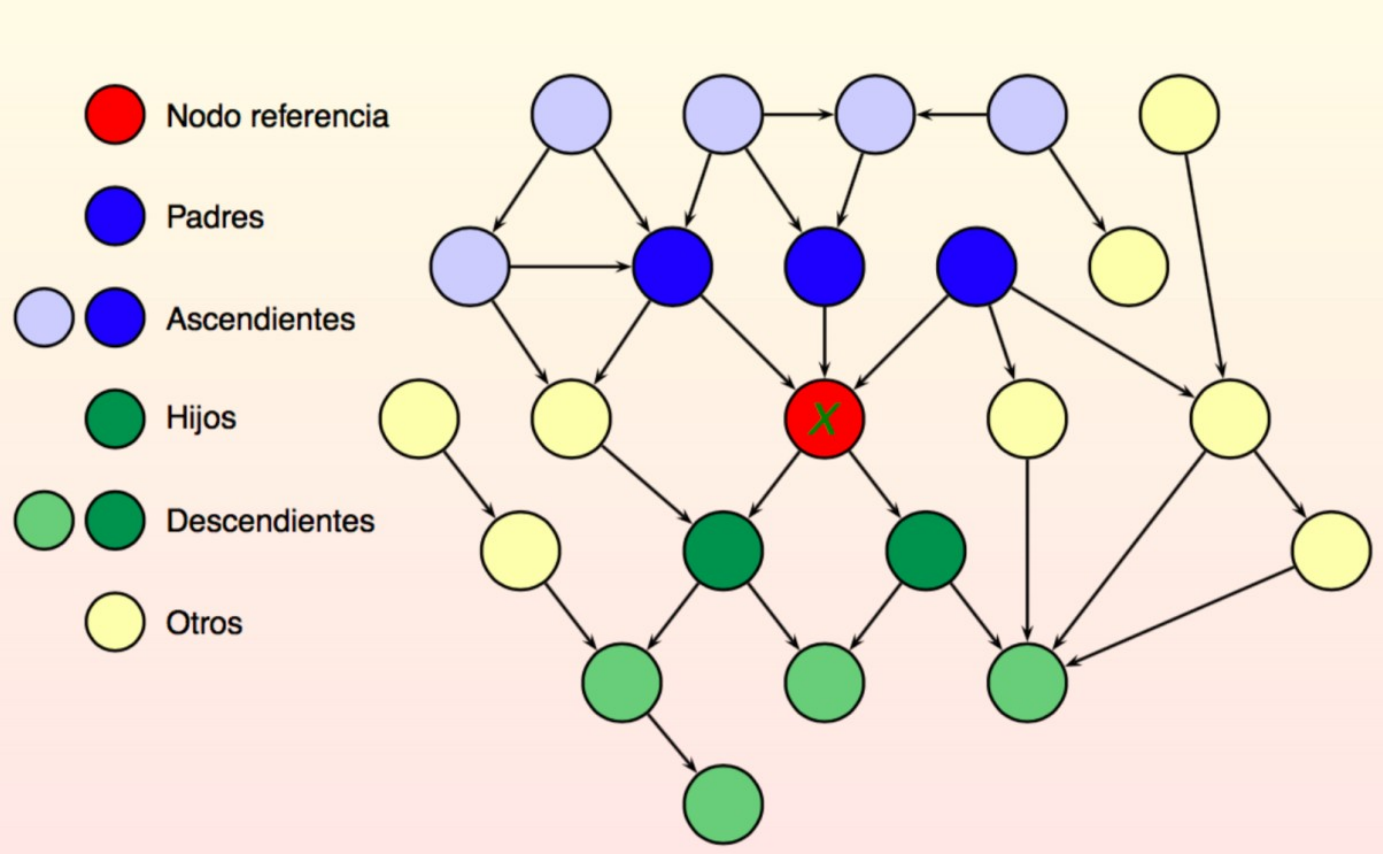

Grafos Dirigidos
Conexos
Inconexos
Acíclicos
Cíclicos
Árboles
Múltiplemente Conexos
Árboles Simples
Poliárboles
A
B
C
D
E
F

Matriz de adyacencia

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$



Grafos


Nodo referencia
Padres
Ascendientes
Hijos
Descendientes
Otros
X

Modelos en Grafos

Modelos en Grafos

- Aprender la estructura

Búsqueda Exhaustiva
Algoritmo PC
Hill-Climbing
Árbol
...

- Aprender los parámetros probabilísticos

Maximum Likelihood
Bayesian
Divergencia KL
...

- Hacer inferencia

Montecarlo
Sampling methods...

Modelos en Grafos

Aprender la estructura

Relacionado con:

- Probabilidad condicionada

$$P(X|Y)$$

- Medidas de independencia

$$I(X,Y)$$

- Medidas de independencia condicionada

$$I(X,Y|Z)$$

- Relaciones Causales

$$X \rightarrow Y$$
$$Y \rightarrow X$$

Aprender la estructura: Algoritmo PC

Asumciones:

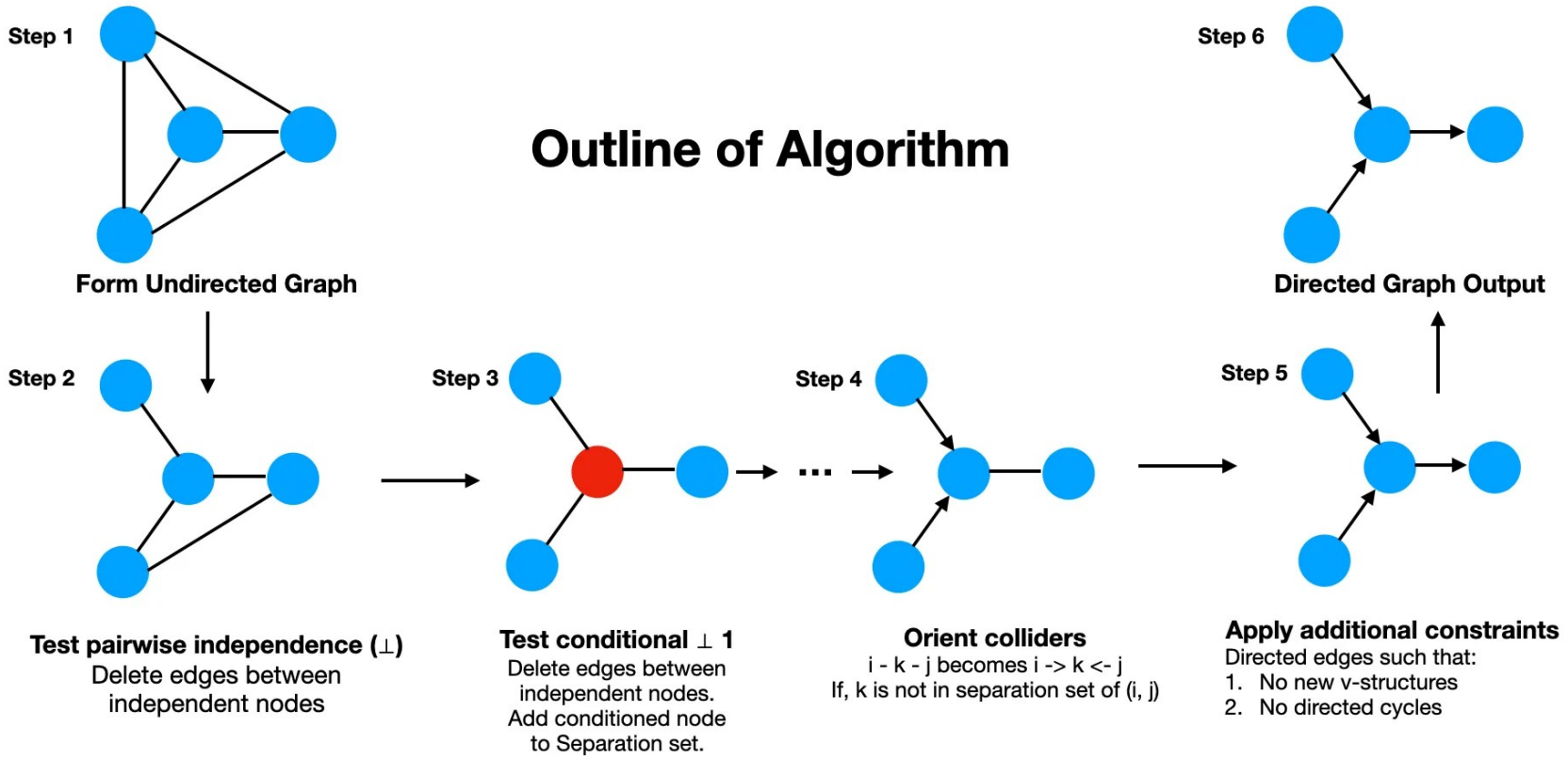
1) ACÍCLICO

2) Propiedad de Markov

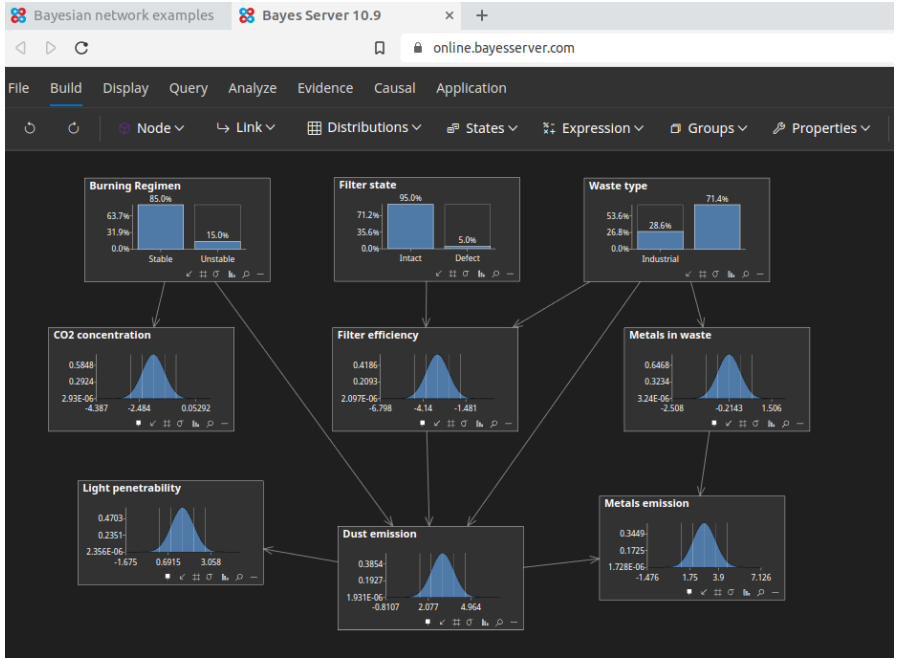
3) Completitud

https://towardsdatascience.com/causal-discovery-6858f9af6dcb

Aprender la estructura: Algoritmo PC

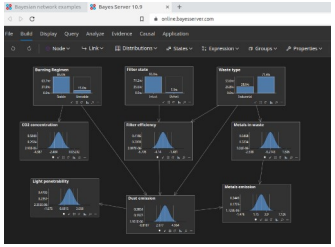


Ejemplos: Bayes server



https://online.bayesserver.com/

https://online.bayesserver.com/



Ejemplos: Bayes server

Con tus datos:

1) Nueva red:
   - File → New

2) Cargar datos (en Excel) :
   - Data explorer → Load data
   - Data connection + Data
   - Information → Edit all → Include = On
   - → Next

3) Cargar Nodos
   - Build → Node → Add Nodes from Data → Next

4) Aprender estructura
   - Link → Structural learning → Algorithm (PC) → Next

5) Aprender Dsitribuciones
   - Distributions → Parameter Learning → Next

Ejemplos: pgmpy

https://pgmpy.org

https://github.com/pgmpy/pgmpy

Librería de Python

Ejemplos: PyBNesian

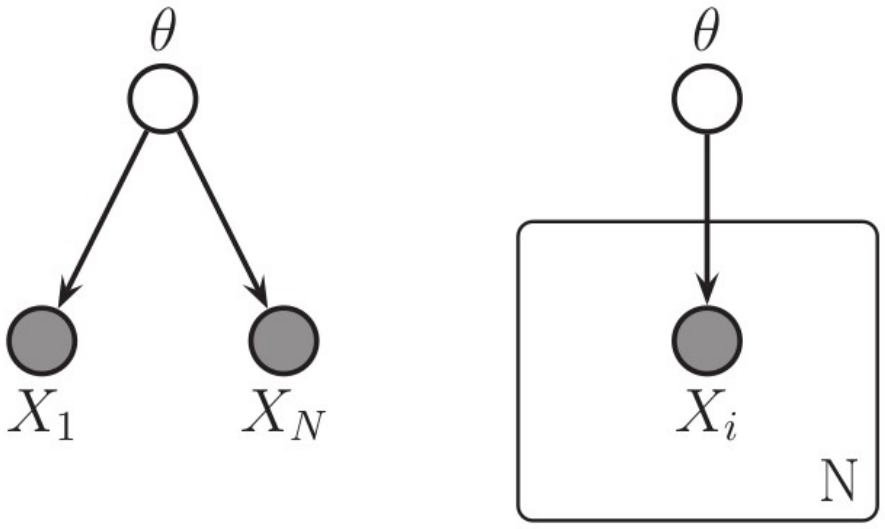
https://github.com/davenza/PyBNesian

David Atienza

Librería de Python

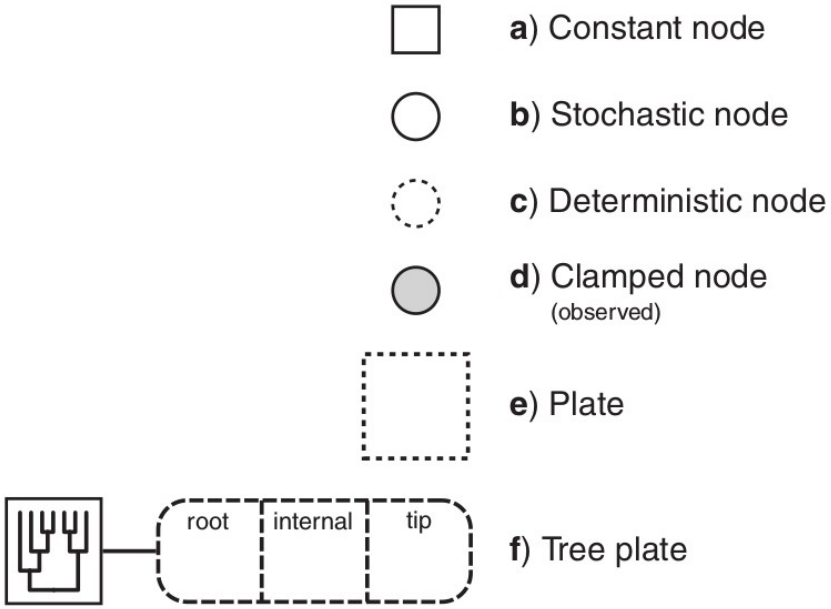
Notación de Modelos en Grafos

Notación



https://revbayes.github.io/tutorials/intro/graph_models.html

Notación



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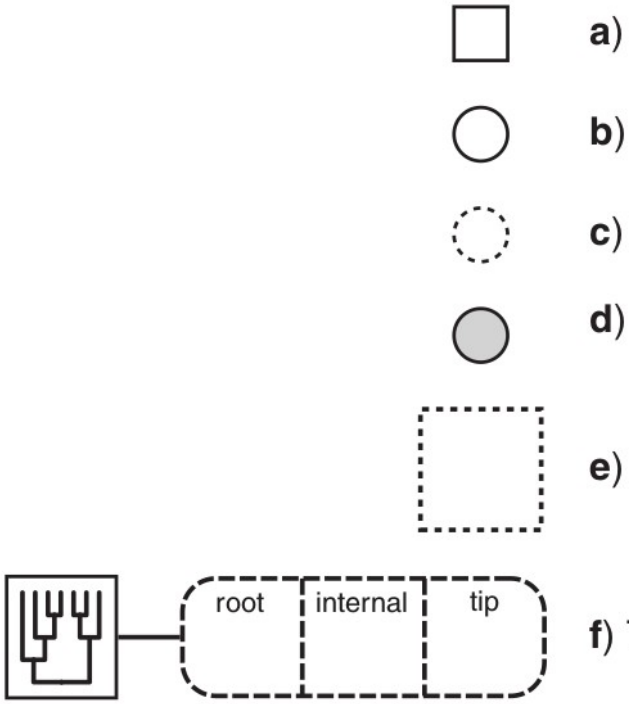
Notación



a) Los **cuadrados sólidos** representan nodos constantes, que especifican variables de valor fijo.

b) Los nodos estocásticos están representados por **círculos sólidos.** Corresponden a variables aleatorias y pueden depender de otras variables.

c) Los nodos deterministas (**círculos punteados**) indican variables determinadas por una función específica aplicada a otra variable. Pueden considerarse transformaciones de variables.

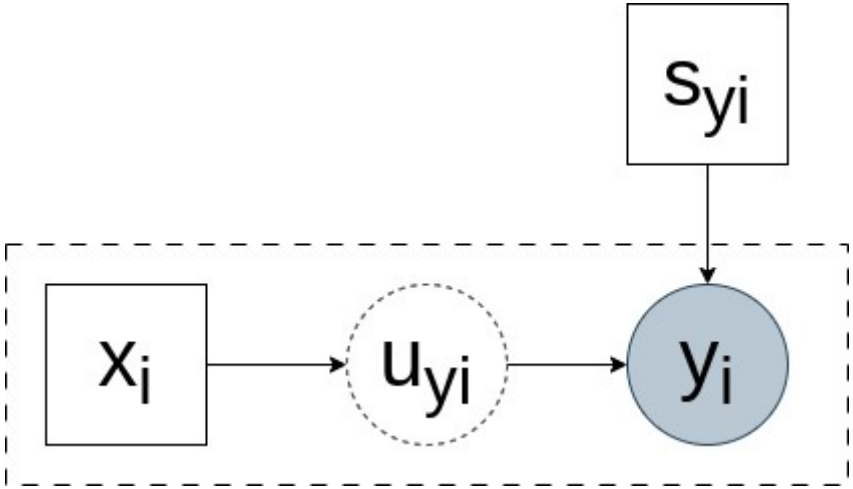
d) Los estados observados se colocan en nodos estocásticos sujetos, representados por **círculos sombreados en gris**.

e) La replicación sobre un conjunto de variables se indica encerrando los nodos replicados en una placa (**rectángulo discontinuo**).

f) Los **árboles** representan las distintas posibles combinaciones.

Notación

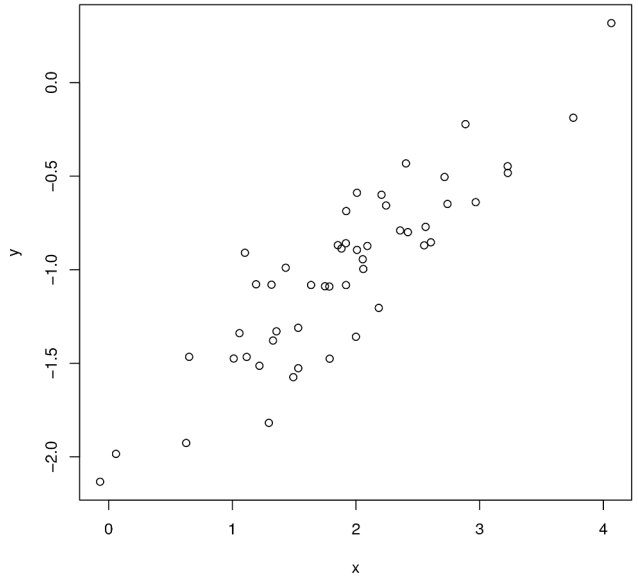
Ejemplos



Draw.io

Notación

Ejemplo: regresión lineal



$$y = \beta x + \alpha + \epsilon$$

$$\mu_y := \beta x + \alpha$$

$$y \sim \mathrm{Normal}(\mu_y, \sigma_\epsilon)$$

Notación

Ejemplo: regresión lineal bayesiana

$$y = \beta x + \alpha + \epsilon$$

$$\mu_y := \beta x + \alpha$$

$$y \sim \text{Normal}(\mu_y, \sigma_\epsilon)$$

$$\beta \sim \text{Normal}(\mu = 0, \sigma^2 = 1)$$

$$\alpha \sim \text{Normal}(\mu = 0, \sigma^2 = 1)$$

$$\sigma_\epsilon \sim \text{Exponential}(\lambda = 1)$$

Notación

Ejemplo: regresión lineal bayesiana

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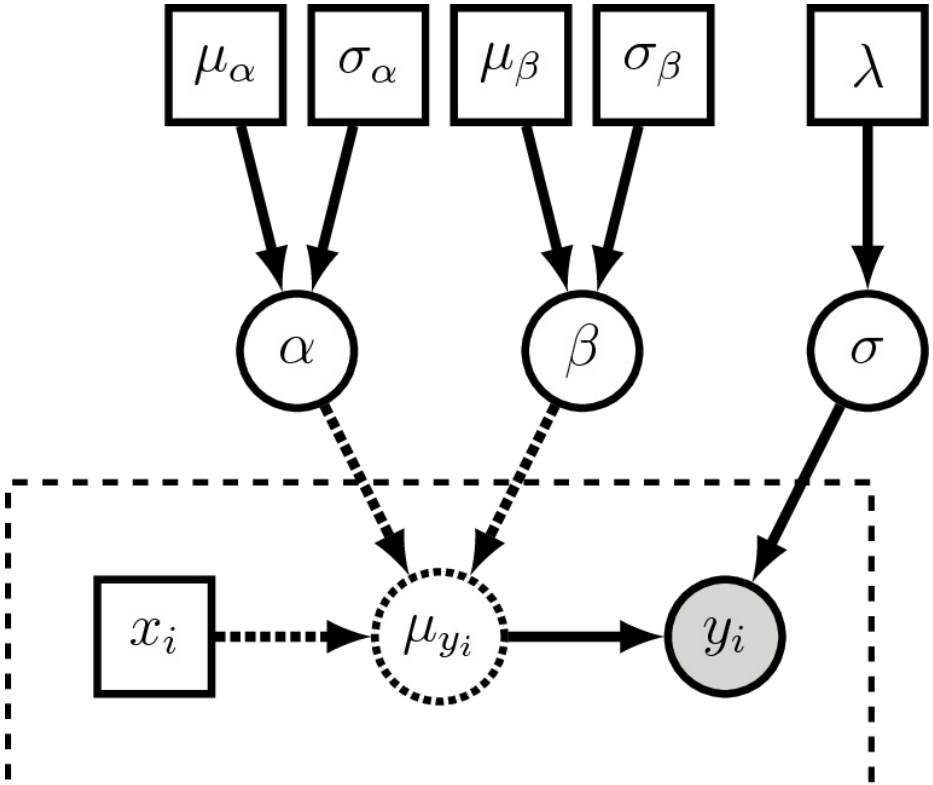


Notación

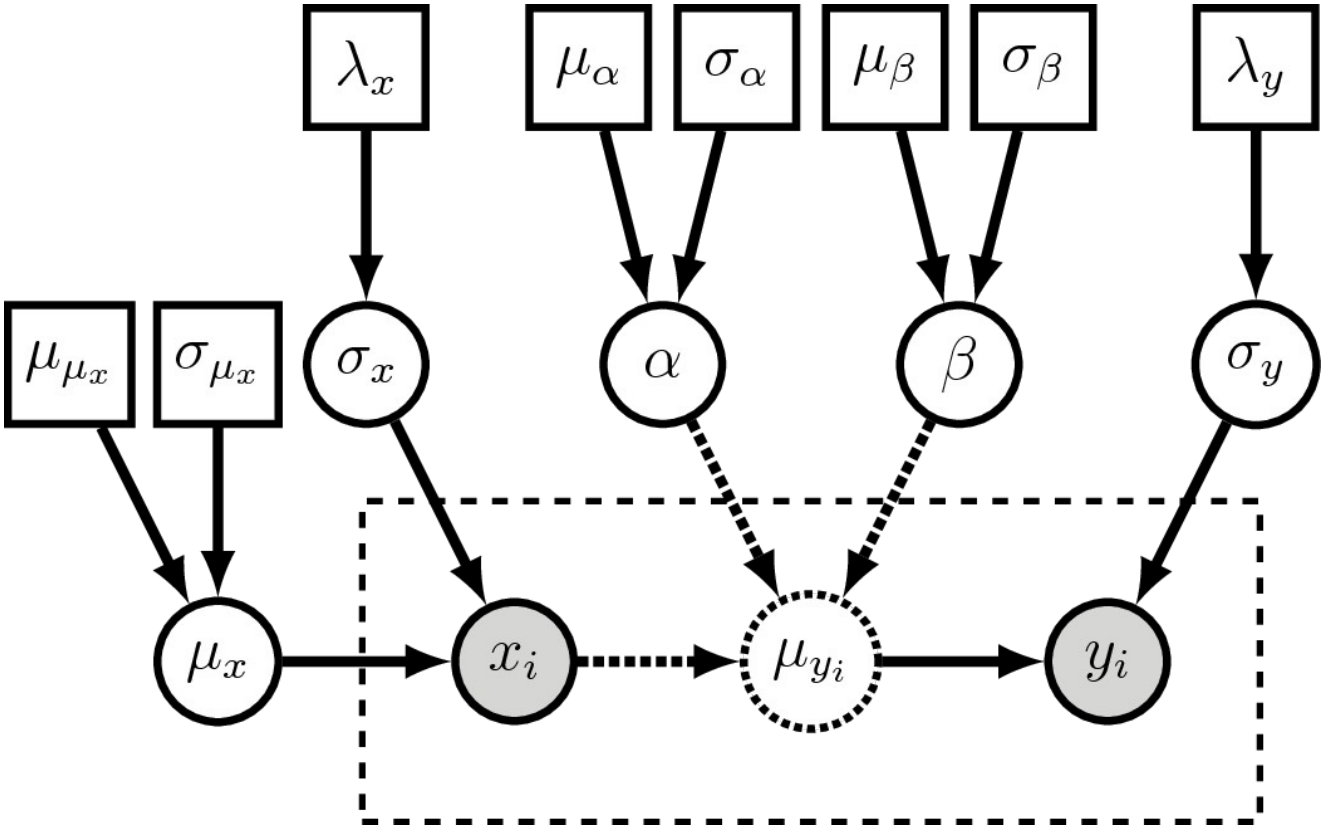
Ejemplos: regresión lineal ?

Notación

Ejemplos: regresión lineal bayesiana con x estocástico?

Notación

Ejemplos: regresión lineal bayesiana con x estocástico?



EXTRA

Libro de Ejercicios “a mano”

Pen & Paper

Exercises in Machine Learning

Michael U. Gutmann

https://arxiv.org/pdf/2206.13446.pdf